

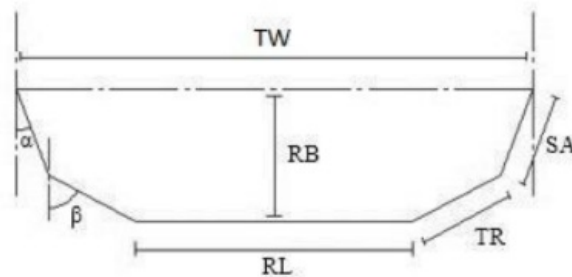
Steering Linkage Optimization for Reduced Driver Effort

Objective

Peak steering torque felt at the driver's hands was identified as a design priority for the current steering linkage. This report documents the optimization methodology used to reduce peak steering torque while preserving Ackermann steering accuracy, and presents the resulting reduction from 6.8 Nm to 4.95 Nm.

1. Steering Linkage Geometry

The steering system was modelled as a 6-bar linkage with one fixed ground link (the trackwidth, defined here as the distance between the ends of the two steering arms, not the outer face-to-face tyre distance).



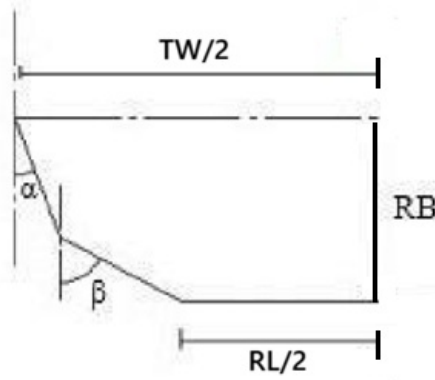
Steering linkage geometry

where:

- **RL** — Rack Length
- **RB** — Rackback
- **TR** — Tierod length
- **SA** — Steering Arm length
- **TW** — Trackwidth (steering-arm-end to steering-arm-end)
- α — angle between the steering arm and vertical, wheels parallel
- β — angle between the tierod and vertical, wheels parallel

Applying the Kutzbach criterion to the full linkage, then adding the physical constraints that the rack remains horizontal and sits at a fixed rackback distance from the trackwidth, reduces the system to a single degree of freedom for a given set of link lengths.

To confirm how many parameters are truly independent, the same criterion was reapplied to one half of the system, treating the rackback length as an additional link:

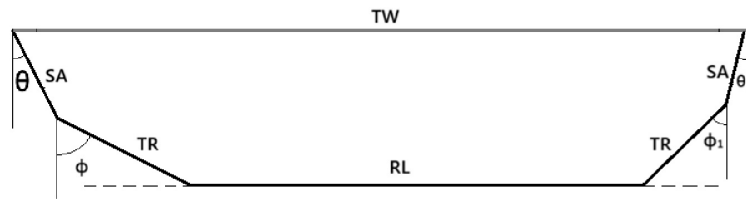


Half-side degrees of freedom

This half-side analysis, combined with the two constraints needed for the geometry to be well defined at the neutral position (rackback-to-trackwidth and rack-to-rackback held at right angles), shows that any four of the seven geometric parameters fully define the linkage. RL, TR, SA, and α were selected as the independent variable set, since they proved most direct to implement in code.

2. Steering Angle Geometry Function

For a given set of link lengths, the inner wheel angle corresponding to any outer wheel angle is derived geometrically rather than read from a table:



Steering angle geometry function

Given an outer wheel angle a , the corresponding linkage angle is:

$$\theta = a + \alpha$$

Two unknowns, ϕ and θ_1 , are then resolved using a vertical constraint (constant rackback length), which holds on both sides of the rack:

$$SA \cdot \cos\theta + TR \cdot \cos\phi = SA \cdot \cos\alpha + TR \cdot \cos\beta \quad (\text{right-hand side known, call it } k1)$$

$$SA \cdot \cos\theta_1 + TR \cdot \cos\phi_1 = SA \cdot \cos\alpha + TR \cdot \cos\beta$$

The first equation gives ϕ directly. A third equation comes from the horizontal constraint, that the horizontal components of both steering arms and tierods, plus the rack length, must sum to the trackwidth:

$$SA \cdot \sin\theta_1 + TR \cdot \sin\phi_1 = TW - RL - SA \cdot \sin\theta - TR \cdot \sin\phi \quad (\text{right-hand side known, call it } k2)$$

ϕ_1 does not need to be solved directly: taking the ϕ_1 terms to the right-hand side in the last two equations, then squaring and adding them, eliminates ϕ_1 (since $\sin^2\phi_1 + \cos^2\phi_1 = 1$) and yields θ_1 directly. The inner wheel steering angle then follows as:

$$\text{inner steering angle} = \alpha - \theta_1$$

Running this function across the full steering range, outer wheel angle from -135° to 135° in 1° steps, produces a 271-point vector of inner wheel angles for any candidate geometry. Comparing this vector against the ideal inner-wheel lookup table (matched to the target Ackermann percentage) gives a 271-element error vector, the quantity every iteration of the optimizer is trying to minimise.

3. Iteration 1 — Unconstrained 4-DOF Optimizer

The first iteration solves purely for Ackermann accuracy. RL, TR, SA, and α are varied by a least-squares solver (`lsqnonlin`) to minimise the inner-wheel angle error vector across the full steering range, starting from hand-calculated initial guesses (to avoid convergence to a non-physical local minimum) and bounded according to packaging constraints from other subsystems.

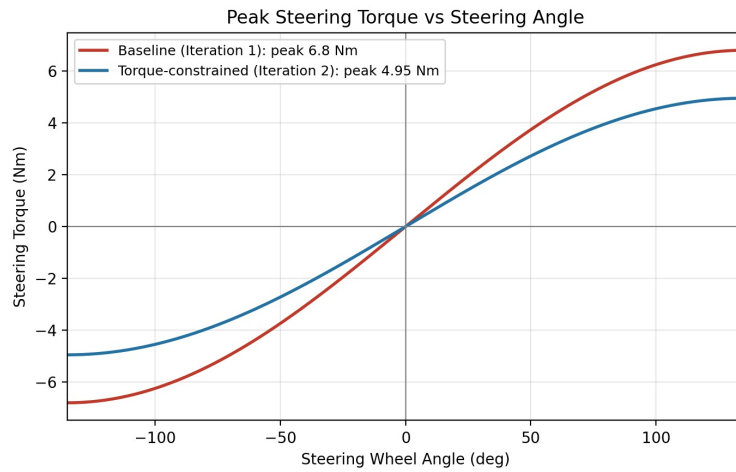
This iteration has no visibility into steering torque. Peak steering torque for the resulting geometry, evaluated separately under a 1.2g lateral and 1.2g longitudinal cornering case across the steering range, was **6.8 Nm**.

4. Iteration 2 — Torque-Constrained Optimizer

The second iteration keeps the same Ackermann-error objective, but adds a nonlinear constraint capping the combined steering torque from both tyres below a defined limit, evaluated under the same 1.2g/1.2g load case across the steering sweep. Because the objective is now constrained rather than unconstrained, the solver was switched from `lsqnonlin` to `fmincon`, which supports general nonlinear constraints (`nonlcon`) alongside the least-squares-style objective.

This directly targets driver effort as a design variable rather than treating it as a byproduct of whatever geometry best matches Ackermann. The resulting geometry converged to a peak steering torque of **4.95 Nm**, a reduction of approximately 27% from the unconstrained baseline, while Ackermann error was checked to confirm it remained within acceptable bounds of the iteration 1 result.

5. Result



Steering torque vs steering angle

This curve is constructed to match the two peak torque values obtained from the optimizer (6.8 Nm baseline, 4.95 Nm torque-constrained) at the $\pm 135^\circ$ steering limits, with zero torque at zero steer. It is illustrative of the shape and magnitude of the reduction rather than a pointwise export of raw simulation output.

Constraining the optimizer directly on steering torque, rather than optimising for Ackermann accuracy alone, reduced peak driver-felt steering torque from 6.8 Nm to 4.95 Nm across the full steering range, without materially compromising the wheel-angle accuracy achieved in the unconstrained baseline.